

ORIGINAL ARTICLE

Blood Pressure Outcomes Among US Veterans Initiating Hypertension Treatment, 2014 to 2023

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BACKGROUND: Amid persistently low blood pressure (BP) control rates and pervasive therapeutic inertia, we evaluated trends in pre- and posttreatment BP before and after 2 major disruptions (the 2017 target-lowering guideline and the COVID-19 pandemic) to assess their impact on early hypertension management.

METHODS: This retrospective cohort study of national Veterans Health Administration data included outpatient adults newly diagnosed with hypertension and starting antihypertensive medication between November 13, 2014, and May 31, 2023 (index date). Patients were stratified into 3 periods corresponding to preguideline/prepandemic (Period 1), postguideline/prepandemic (Period 2), and postguideline/postpandemic (Period 3), and into 3 pretreatment systolic BP (SBP) groups based on the average of ≥ 2 measurements in the 90-day preindex period (<140 , 140–160, or ≥ 160 mm Hg). Across periods, multivariable analyses evaluated: (1) mean posttreatment SBP (mean of measurements from days 180–365 postindex); and (2) posttreatment BP control ($<140/90$ or $<130/80$ mm Hg).

RESULTS: Among 271 496 Veterans (mean age, 62 years; 92% male; 66% non-Hispanic White), 39%, 32%, and 29% were in Periods 1, 2, and 3, respectively. Adjusted posttreatment SBP across periods was 128, 135, and 142 mm Hg for the <140 , 140 to 160, and ≥ 160 groups, respectively. Rates of posttreatment BP control $<140/90$ mm Hg across Periods 1, 2, and 3, respectively, were 82.3%, 83.3%, and 84.2% (SBP <140 group); 64.1%, 66.1%, and 67.3% (140–160 group); and 46.4%, 47.0%, and 48.6% (≥ 160 group). For BP control $<130/80$ mm Hg, rates were 38.2%, 40.0%, and 39.9% (SBP <140 group); 20.7%, 22.0%, and 22.6% (140–160 group); and 15.1%, 15.4%, and 15.2% (≥ 160 group).

CONCLUSIONS: Despite the target-lowering guideline and the care-disrupting pandemic, BP levels and control among Veterans remained largely unchanged. At 1 year, only half to two-thirds achieved BP $<140/90$ mm Hg, and few reached $<130/80$ mm Hg, underscoring persistent clinical inertia and the need to improve early hypertension management in the Veterans Health Administration.

Key Words: adult ■ hypertension ■ middle aged ■ outpatients ■ pandemics

For decades, the Veterans Health Administration (VHA) has exemplified hypertension care, blending evidence-based treatment with system-wide implementation strategies that pushed blood pressure (BP) control rates to $>75\%$ at select facilities in the early 2000s.¹ Yet, after US surveillance data signaled a decrease in population-level BP control starting in 2014,^{2,3} there has been scant

evidence describing BP control rates among Veterans who newly start antihypertensive therapy, particularly with respect to the stricter targets recommended by contemporary clinical guidelines.

Two major, sequential events may have impacted BP control in the VHA. First, the 2017 American College of Cardiology/American Heart Association (ACC/AHA)

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WHAT IS KNOWN

- Blood pressure (BP) control rates among US adults have declined since ≈2014, and clinical inertia occurs in up to half of hypertension-related clinical encounters nationwide.
- The 2017 American College of Cardiology/American Heart Association guideline lowered the recommended threshold for antihypertensive treatment initiation and intensification from ≥140/90 mm Hg to ≥130/80 mm Hg for adults at elevated cardiovascular risk, yet national BP control rates did not meaningfully improve following the guideline publication.
- The COVID-19 pandemic disrupted hypertension care delivery, with studies in the general population reporting modest decreases in BP control during the acute pandemic period; however, comparable data specifically among Veterans initiating antihypertensive medication—using a new-user study design with separate evaluation of guideline and pandemic impacts—have been lacking.

WHAT THE STUDY ADDS

- Among 271 496 US Veterans newly initiating antihypertensive medication between 2014 and 2023, pretreatment BP levels — a marker of when clinicians decided to act—did not meaningfully change before or after the 2017 American College of Cardiology/American Heart Association guideline or the onset of the COVID-19 pandemic, indicating persistent clinical inertia across all 3 time periods studied.
- Despite the guideline-recommended target of <130/80 mm Hg, only two-thirds of Veterans with stage 2 hypertension (pretreatment SBP ≥140 mm Hg) achieved BP control to <140/90 mm Hg within 1 year of treatment initiation, and fewer than 1 in 4 achieved the more stringent <130/80 mm Hg threshold—with these rates remaining largely static across periods.
- BP control gaps were disproportionately borne by Veterans who were female or from racial and ethnic minority backgrounds; notably, non-Hispanic Black Veterans in the 140–160 mm Hg stratum and Hispanic and female Veterans with severe hypertension (SBP ≥160 mm Hg) experienced declining BP control rates in the postpandemic period, underscoring the need for equity-focused, multilevel interventions to improve early hypertension management in the Veterans Health Administration.

BP guideline lowered the threshold for antihypertensive medication initiation and intensification from systolic BP (SBP) ≥140 mm Hg or diastolic BP (DBP) ≥90 mm Hg to SBP/DBP of 130/80 mm Hg for US adults at cardiovascular risk.⁴ Second, the COVID-19 pandemic disrupted clinical care in 2020, reshaping how Americans received healthcare.⁵ While several studies reported modest reductions in BP control in the general

Nonstandard Abbreviations and Acronyms

ACC	American College of Cardiology
AHA	American Heart Association
BP	blood pressure
DBP	diastolic blood pressure
PDC	proportion of days covered
SBP	systolic blood pressure
VHA	Veterans Health Administration

population during these times,^{6–8} similar VHA studies have included mixed hypertensive populations (treated and untreated alike) or prevalent antihypertensive medication users, without separately examining the impacts of each event on BP control among new treatment initiators.^{9,10}

Clinical inertia, the failure to start or intensify treatment despite evidence of uncontrolled disease, occurs in 30% to 55% of hypertension-related visits and is a central barrier to achieving BP control nationwide.^{11–17} Factors contributing to inertia, such as pretreatment BP level and reliance on monotherapy, may also influence medication adherence, further limiting treatment effectiveness.¹⁸ Understanding BP management in patients newly initiating therapy, who may be especially vulnerable to inertia during periods of change or care disruption, is particularly important because early treatment response strongly influences long-term BP outcomes and cardiovascular risk.^{19–23} Prior work has evaluated upstream gaps in rates of hypertension awareness, diagnosis, and treatment in the VA.^{9,24,25} To further inform the VA's future efforts and policies to advance BP control, we sought to evaluate early management among Veterans newly initiating antihypertensive medications using a new-user study design. We evaluated differences in pre- and posttreatment BP levels and 1-year BP control and medication adherence in 3 time periods (preguideline/prepandemic, postguideline/prepandemic, and postguideline/postpandemic). Therefore, we conducted the following analysis to assess differences in pre- and posttreatment BP levels and 1-year BP control and medication adherence among Veterans initiating treatment for hypertension in 3 time periods (preguideline/prepandemic, postguideline/prepandemic, and postguideline/postpandemic). Therapeutic inertia encompasses both failure to initiate and failure to intensify therapy; this study focuses on postinitiation inertia by evaluating early management and outcomes during the first year after treatment initiation. By providing a current benchmark near the publication of the 2025 AHA/ACC BP Guideline,²⁶ these findings aim to inform the VHA's future efforts and policies to advance BP control for all Veterans.

METHODS

Data Sources and Oversight

We used data from the VHA Corporate Data Warehouse, which contains information back to 1999 on outpatient and inpatient clinical encounters, patient sociodemographic information, vital signs, laboratory test results, diagnosis codes, and medications filled through VHA pharmacies. The University of Utah institutional review board and Salt Lake City Veterans Affairs Health Care System Research and Development Office approved the current study with a waiver of informed consent. We followed the STROBE reporting guideline for cohort studies (Table S1).²⁷

Study Design and Population

We conducted a new-user cohort study of Veterans who filled a prescription for antihypertensive medication in the VHA outpatient setting from November 13, 2014, through May 31, 2023 (ie, the index date identification period). The date of each Veteran's first prescription fill for an oral antihypertensive medication during this period was defined as the index date (N=4699410; Figure S1). We further excluded Veterans who: (1) did not have continuous VHA coverage for at least 2 years before the index date (N=876989); (2) did not have ≥ 1 inpatient or ≥ 2 outpatient encounters during the pretreatment period with a hypertension diagnosis code (401.x, 403.0x, 403.1x, 403.9x; I10, I12.0, I12.9.9) in the 1 year before, or 180 days after, the index date (N=911108)²⁸; (3) filled an antihypertensive medication before the index date (N=2423299); (4) did not have 2 valid BP measurements in the 1 year before the index date (N=214618); (5) were < 18 years of age or pregnant on the index date (N=39); and (6) were marked as non-Veterans or had a death date before the index date (N=1861). The final sample included 271496 Veterans.

Data Collection

We collected data on baseline (ie, preindex) characteristics from the VHA Corporate Data Warehouse using procedure and diagnosis codes associated with inpatient and outpatient encounters on or before the index date (detailed definitions in Table S2). As treatment recommendations and patterns vary by race and ethnicity, self-reported race and ethnicity were queried from enrollment files and grouped into 5 mutually exclusive categories: Hispanic, non-Hispanic Black, non-Hispanic White, non-Hispanic Other (inclusive of multiracial, Asian American, American Indian or Alaska Native, and Native Hawaiian or Other Pacific Islander), and missing (defined as missing ethnicity or non-Hispanic with missing race). We also defined a variable indicating the time interval between hypertension diagnosis and the index date, defined as the difference in years between the index date and the first inpatient or outpatient hypertension diagnosis code during the study period.

Exposure

Our research question was whether BP outcomes varied over time, with a focus on 2 key events: the 2017 guideline change and the onset of the COVID-19 pandemic. To characterize the exposure, we first evaluated annualized trends in pre- and posttreatment systolic BP from 2014 to 2023. This allowed visualization of the acute COVID-19 disruption (2020–2021)

and subsequent stabilization (2022–2023). Annualized mean BP values (Figure S2) were unchanged from 2020 to 2021, followed by re-established stability through 2023, indicating that aggregation of the postpandemic period would not obscure meaningful within-period differences. Therefore, we selected a 3-level exposure variable to streamline the analysis and align with the study aims. Periods were defined by index date (Figure 1A): (1) preguideline/prepandemic (November 13, 2014–November 30, 2017; N=104879), (2) postguideline/prepandemic (December 1, 2017–December 31, 2019; N=86230), and (3) postguideline/postpandemic onset (January 1, 2020–May 31, 2023; N=80387).

Study Outcomes

The 2 coprimary outcomes were mean pretreatment SBP levels and mean 1-year posttreatment SBP levels. We secondarily evaluated posttreatment BP control dichotomously at 2 levels: $< 140/90$ mmHg (JNC-8 threshold, which is still used for Health Effectiveness Data Information Set performance metrics) and $< 130/80$ mmHg (2017 ACC/AHA threshold). We also evaluated the proportion achieving the individual SBP and DBP thresholds separately and evaluated an SBP goal of < 150 mmHg for individuals ≥ 60 years of age (American College of Physicians/American Academy of Family Physicians 2017 threshold²⁹). For SBP and DBP, we selected valid BP measurements taken in outpatient settings that typically manage hypertension (Table S2).^{1,30} Pretreatment BP was defined as the mean of all valid BP measurements, measured on ≥ 2 separate dates, within 365 days preindex, and posttreatment BP was defined as the mean of all BP measurements in days 180 through 365 following the index date (Figure 1B).

Because medication adherence influences BP outcomes and clinical inertia, we secondarily assessed 1-year antihypertensive medication adherence using proportion of days covered (PDC) from the index to 1 year postindex.^{31,32} We separately defined PDC continuously and categorically at a threshold of 80%, as recommended by the Pharmacy Quality Alliance³³ and other studies.^{28,34}

The administrative end of follow-up was November 2024; longer horizons would preferentially exclude more recent initiators and rely on outcome elements (eg, death) with greater latency near the cutoff. These analyses focused on early management because guidelines during the study period recommended monthly reassessment/titration until control.^{4,35}

Statistical Analysis

Descriptive Statistics

Characteristics of the population were summarized overall and by period using descriptive statistics (Supplemental Methods). Because treatment initiation depends on pretreatment SBP levels, we assessed outcomes within strata of pretreatment SBP (< 140 , 140–160, and ≥ 160 mmHg) for each period. Patients missing baseline values for the continuous variables of total cholesterol, HDL cholesterol, or eGFR (Table S3) were excluded from the corresponding summary statistics. For dichotomous variables, those missing baseline data were assumed not to have the condition of interest, although we created a missing indicator for patients with missing ethnicity or non-Hispanic patients with missing race data.

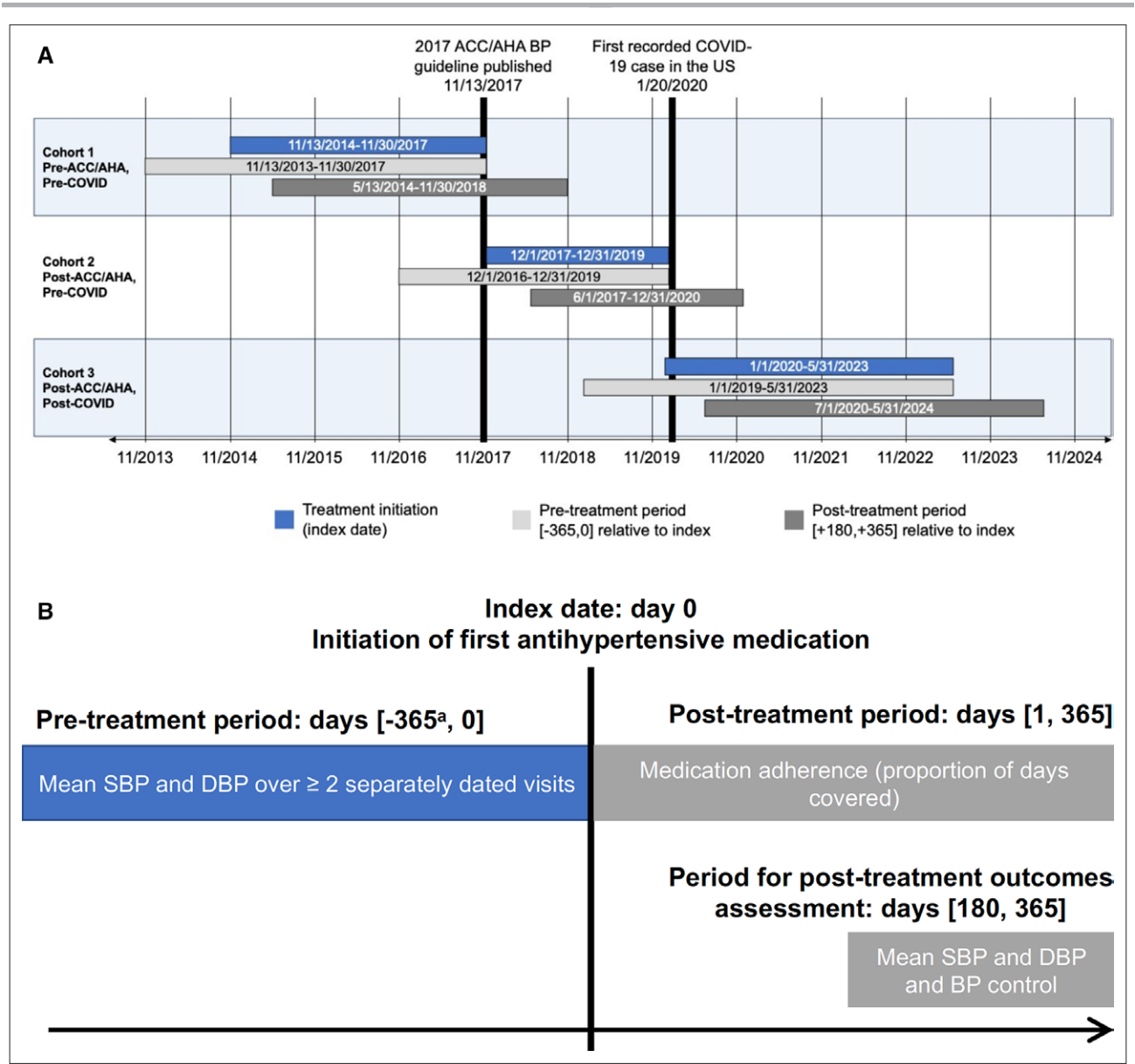


Figure 1. Timeline of the periods and study schema. **A**, Timeline. **B**, Patient-level assessment of pretreatment and posttreatment period. ^aIf ≥2 blood pressure (BP) measurements were recorded up to 90 days before the index date, then pretreatment blood pressure was determined using the mean of these measurements. Otherwise, the number of lookback days was increased up to 365 to obtain 2 blood pressure measurements. ACC indicates American College of Cardiology; AHA, American Heart Association; COVID-19, coronavirus disease 2019; DBP, diastolic blood pressure; SBP, systolic blood pressure; and US United States.

Weighted Outcome Models

To estimate period-specific outcomes, linear regression modeled continuous posttreatment SBP and DBP levels, quasibinomial regression modeled the continuous PDC outcome, and Poisson regression with robust error variance modeled binary outcomes. We generated 2 weights using the weightit package: (1) inverse probability of treatment weights to adjust for confounding covariates and potential risk factors for the outcomes; and (2) dropout weights to address potential bias due to missing posttreatment BP outcomes (Table S4).³⁶ Both weights incorporated sex, race, ethnicity, diabetes, chronic kidney disease, and history of cardiovascular disease. Dropout weights were estimated in the entire sample, then outcome

models were weighted and restricted to those with nonmissing postindex BP values (N=188 705; 69.5%). For medication adherence, outcome models included all Veterans regardless of postindex BP availability. The product of the 2 weights was used in the regression models, in addition to a categorical variable indicating the calendar month of the index date to account for potential seasonal variation in SBP.³⁷ Standard methods were used to assess covariate balance before and after weighting (Supplemental Methods).³⁸

To estimate period-specific means, proportions, and 95% CIs, we used model-based marginal estimates derived from the weighted generalized linear models using the emmeans package. We conducted pairwise comparisons between

periods using adjusted contrasts of the marginal estimates and reported 2-sided *P* values for the comparisons between Period 1 versus Period 2 and Period 2 versus Period 3, with $P < 0.05$ considered statistically significant (Tukey method adjusted for multiple comparisons to maintain the family-wise error rate). The analyses reflect early postinitiation management (1-year BP and adherence), a direct indicator of intensification and support rather than simple initiation rates.

Subgroup and Sensitivity Analyses

For comparability with prior VHA BP studies,^{1,9,10,39} we evaluated mean posttreatment SBP and SBP control in the overall cohort, not stratified by pretreatment SBP groups. We also assessed subgroups by age, sex, race and ethnicity, diabetes, cardiovascular disease, CKD, and index combination therapy use. We also conducted a descriptive analysis evaluating overall combined SBP/DBP control by VISN, and conducted a post hoc sensitivity analysis incorporating VISN as a random effect in the models. Although the primary analysis accounts for inherent differences between patients with and without follow-up BP values using dropout weights, we performed a sensitivity analysis without the dropout weights, similar to previously published studies.^{40,41} Finally, because patients may be seen only once annually, or may receive a diagnosis and treatment in 1 visit, we repeated analyses among a cohort requiring only 1 valid BP measurement preindex instead of 2.

Analyses were conducted in R (version 4.4.1).

RESULTS

Baseline Characteristics

There were 271 496 Veterans in the current analysis, with a mean (SD) age of 62 (14) years, 8.4% were female, and 65.8% were non-Hispanic White (Table). The median (interquartile range) number of years between first hypertension diagnosis and the index date across Periods 1, 2, and 3 was 0.8 (interquartile range, 0.0–3.1), 1.0 (0.0–3.7), and 1.0 (interquartile range, 0.0–4.9), respectively (Figure S3 for histograms). Over time, across Periods 1 through 3, Veterans' average age increased, the proportion who were current smokers decreased, total cholesterol and estimated glomerular filtration rate decreased, and the proportion with histories of CKD or cardiovascular disease increased. Initiation of antihypertensive combination therapy decreased, and among those starting combination therapy, the proportion using a fixed-dose combination product also decreased. Weighting adequately balanced the periods in terms of baseline characteristics, with all absolute standardized mean differences < 0.1 (Figure S4).

Primary Outcome: Pre- and Posttreatment BP Levels

In the < 140 mmHg group, the adjusted mean (95% CI) pretreatment SBP levels were 128.7 (128.7–128.8), 128.7 (128.6–128.7), and 128.8 (128.7–128.9) mmHg in Periods 1, 2, and 3, respectively ($P = 0.01$

across periods; Figure 2A; Figure S5A for box plots). In the 140–160 mmHg group, the corresponding values were 147.0 (146.6–147.0), 146.9 (146.9–147.0), and 147.1 (147.1–147.2) mmHg ($P < 0.001$ across periods). For those with SBP ≥ 160 mmHg, mean (95% CI) pretreatment SBP levels were nearly identical across periods at 167.1 (166.9–167.3), 167.1 (166.9–167.3), and 167.0 (166.8–167.2) mmHg ($P = 0.76$ across periods). Adjusted mean pretreatment DBP levels decreased across periods (both $P < 0.001$; Figure S6A).

For mean posttreatment SBP, in the < 140 mmHg stratum, period-specific means (95% CI) were 128.3 (128.2–128.4), 128.1 (128.0–128.3), and 128.0 (127.9–128.2) mmHg ($P = 0.05$); in the 140–160 mmHg group: 135.4 (135.3–135.6), 135.3 (135.1–135.5), and 135.0 (134.9–135.2) mmHg ($P = 0.002$ across periods), whereas in the ≥ 160 mmHg group: 142.9 (142.4–143.4), 142.8 (142.1–143.4), and 142.3 (141.7–142.9) mmHg ($P = 0.30$ across periods; Figure 2B). Within all pretreatment SBP strata, mean posttreatment DBP decreased between Periods 1 and 2 ($P < 0.05$) but was similar between Periods 2 and 3 ($P > 0.05$; Figure S5B for box plots; Figure S6B).



One-Year Posttreatment BP Control

When examining BP control to $< 140/90$ mmHg, control rates (95% CI) were highest among those with pretreatment SBP < 140 mmHg and increased from 82.3% (82.0%–82.7%) in Period 1 to 83.3% (82.9%–83.7%) and 84.2% (83.8%–84.7%) in Periods 2 and 3, respectively (both $P < 0.001$; Figure 3A). In the 140–160 mmHg stratum, rates rose from 64.1% (63.7%–64.5%) to 66.1% (65.6%–66.5%) and 67.3% (66.9%–67.8%) (both $P < 0.001$). Among Veterans with pretreatment SBP ≥ 160 mmHg, control rates across periods were 45.6% (44.7%–46.5%), 46.6% (45.5%–47.7%), and 47.7% (46.6%–48.8%), respectively ($P = 0.60$ for Period 2 versus Period 1; $P = 0.08$ for Period 3 versus Period 2).

For the more stringent $< 130/80$ mmHg threshold, among those with pretreatment SBP < 140 mmHg, control rates (95% CI) were 38.2% (38.0%–38.4%), 40.0% (39.7%–40.3%), and 39.9% (39.6%–40.1%) ($P < 0.001$ for Period 2 versus Period 1; $P = 0.82$ for Period 3 versus Period 2; Figure 3B). In the 140–160 mmHg group, control increased from 20.7% (20.4%, 20.9%) to 22.0% (21.7%–22.2%; $P < 0.001$), then to 22.6% (20.4%–20.9%; $P = 0.002$). Among those with SBP ≥ 160 mmHg, control across periods was 15.1% (14.6%–15.6%), 15.4% (14.8%–16.0%), and 15.2% (14.6%–15.8%) ($P = 0.83$ for Period 2 versus Period 1; $P = 0.94$ for Period 3 versus Period 2). BP control at individual SBP and DBP thresholds, separately, is shown in Table S5. Among 115 704 Veterans ≥ 60 years of age (61.3% of the overall period), depending on pretreatment strata, posttreatment SBP control to < 150 mmHg ranged from 68.2%

Table. Characteristics of Veterans With Hypertension Who Initiated Antihypertensive Medication, 2014 to 2023

Characteristics	Overall, N=271 496	Period*		
		Period 1	Period 2	Period 3
		N=104 879	N=86 230	N=80 387
Age, y				
Mean (SD)	62 (14)	61 (13)	62 (14)	63 (14)
Median (25th–75th percentile)	64 (53–71)	63 (52–69)	64 (53–72)	64 (54–74)
Age category, n (%)				
18–44, n (%)	33 353 (12.3)	13 271 (12.7)	10 508 (12.2)	9574 (11.9)
45–64, n (%)	112 478 (41.4)	45 471 (43.4)	35 019 (40.6)	31 988 (39.8)
65–74, n (%)	84 947 (31.3)	33 831 (32.3)	28 033 (32.5)	23 083 (28.7)
≥75, n (%)	40 718 (15.0)	12 306 (11.7)	12 670 (14.7)	15 742 (19.6)
Female, n (%)	22 735 (8.4)	8216 (7.8)	7063 (8.2)	7456 (9.3)
Race and ethnicity group, n (%)				
Hispanic	17 045 (6.3)	6352 (6.1)	5467 (6.3)	5226 (6.5)
Non-Hispanic Black	49 836 (18.4)	19 329 (18.4%)	15 954 (18.5%)	14 553 (18.1%)
Non-Hispanic White	178 583 (65.8)	69 703 (66.5%)	56 505 (65.5%)	52 375 (65.2%)
Non-Hispanic Other†	8626 (3.2)	3315 (3.2%)	2717 (3.2%)	2594 (3.2%)
Missing ethnicity or non-Hispanic and missing race	17 406 (6.4)	6180 (5.9%)	5587 (6.5%)	5639 (7.0%)
Current smoker, n (%)	48 374 (17.8)	20 950 (20.0)	14 052 (16.3)	13 372 (16.6)
Time interval between first hypertension diagnosis code and index date, y, median (25th–75th percentile)	0.9 (0.0–3.)	0.8 (0.0–3.1)	1.0 (0.0–3.7)	1.0 (0.0–4.9)
Pretreatment systolic blood pressure category, n (%)				
<140 mm Hg	153 744 (56.6)	59 713 (56.9)	49 623 (57.5)	44 408 (55.2)
140 to <160 mm Hg	103 577 (38.2)	39 713 (37.9)	32 280 (37.4)	31 584 (39.3)
≥160 mm Hg	14 175 (5.2)	5453 (5.2)	4327 (5.0)	4395 (5.5)
Total cholesterol, mg/dL, median (25th–75th percentile)	179 (152–208)	181 (155–210)	178 (151–207)	178 (150–207)
HDL cholesterol, mg/dL, median (25th–75th percentile)	44 (37–54)	44 (37–53)	44 (37–54)	45 (38–54)
Estimated glomerular filtration rate, mL/min per 1.73 m ² , mean (SD)	83 (19)	84 (19)	83 (19)	82 (19)
Diabetes, n (%)	68 846 (25.4)	25 901 (24.7)	21 876 (25.4)	21 069 (26.2)
Chronic kidney disease, n (%)	43 246 (15.9)	14 550 (13.9)	14 005 (16.2)	14 691 (18.3)
History of CVD, n (%)	45 305 (16.7)	16 536 (15.8)	14 577 (16.9)	14 192 (17.7)
Combination therapy‡, n (%)	47 638 (17.5)	19 902 (19.0)	15 511 (18.0)	12 225 (15.2)
Initiated ≥1 single-pill combination product§, n (%)	18 120 (38.0)	7956 (40.0)	5756 (37.2)	4408 (36.1)

$P<0.001$ for each characteristic across periods. For variables with missing values (race, total cholesterol, and cardiovascular disease risk), denominators to calculate percentages were determined by the number of Veterans with nonmissing values. Values are mean (SD) for continuous and frequency (%) for categorical variables, unless otherwise specified. ACC indicates American College of Cardiology; AHA, American Heart Association; CVD, cardiovascular disease; and HDL, high-density lipoprotein.

*Period 1: pre-ACC/AHA guideline November 2014 to November 30, 2017; Period 2: post-ACC/AHA guideline, pre-COVID January 12, 2017–December 12, 2019; post-ACC/AHA guideline, post-COVID January 1, 2020 to May 31, 2023.

†Includes multiracial, Asian American, American Indian or Alaskan Native, and Native Hawaiian or Other Pacific Islander.

‡Defined as initiating ≥2 antihypertensive medication classes.

§Denominator is the combination therapy N from the preceding row.

to 95.3%, with no significant differences detected across periods ($P>0.05$; Figure S7).

Twelve-Month Medication Adherence

In each pretreatment SBP period, mean PDC and the proportion with PDC ≥80% increased across periods (Table S6). For example, among Veterans with pretreatment SBP 140–160 mm Hg, mean PDC increased from 67.7% (67.4%–68.0%) to 69.2% (68.9%–69.5%;

$P<0.001$), then to 70.8% (70.5%–71.1%; $P<0.001$), with corresponding increases in the PDC ≥80% from 47.3% (46.9%–47.7%) to 49.4% (48.9%–49.8%; $P<0.001$), then to 51.5% (51.0%–52.0%; $P<0.001$).

Subgroup and Sensitivity Analyses

When combining all SBP strata, the mean (95% CI) posttreatment SBP was 131.7 (131.6–131.8), 131.5 (131.4–131.6), and 131.6 (131.4–131.8) mm Hg in

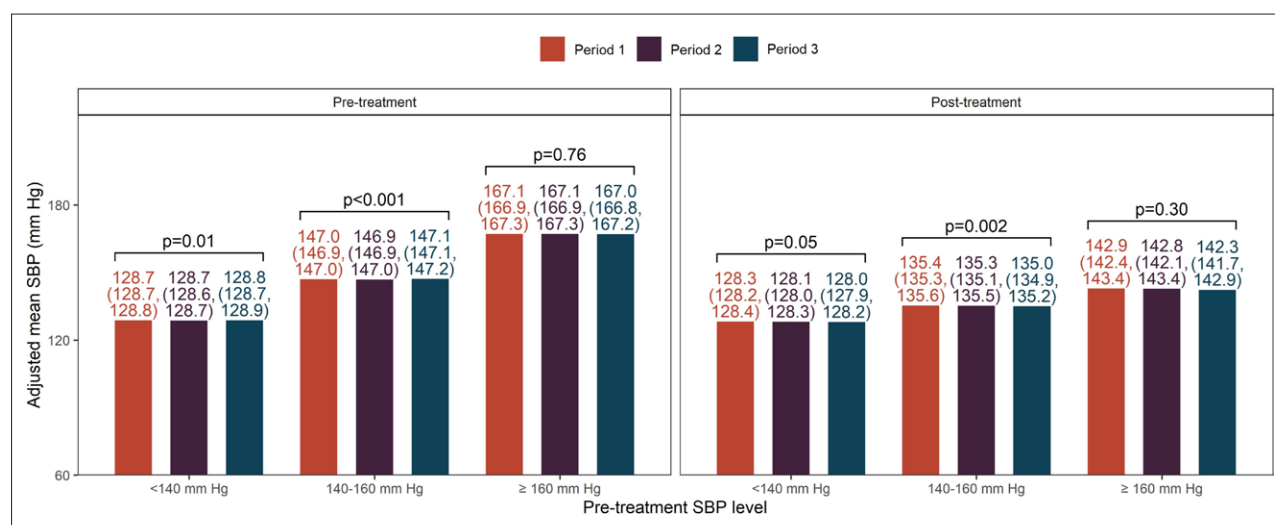


Figure 2. Adjusted mean pre- and posttreatment systolic blood pressure (SBP) levels by period.

The figure shows adjusted mean pre- and posttreatment SBP values in each period, with 95% CIs contained in parentheses. Values are derived from models adjusting for calendar month, age, sex, race, ethnicity, diabetes, chronic kidney disease, and history of cardiovascular disease.

Periods 1, 2, and 3 ($P=0.04$ for Period 2 versus Period 1; $P=0.83$ for Period 3 versus Period 2; [Table S7](#)). BP control to <140/90 mm Hg was 73.7% (73.4%–73.9%), 75.1% (74.8%–75.4%), and 75.7% (75.4%–76.0%) across Periods 1, 2, and 3, respectively ($P<0.001$ for Period 2 versus Period 1; $P=0.02$ for Period 3 versus Period 2), and corresponding values for the <130/80 mm Hg threshold were 30.5% (30.3%–30.6%), 32.1% (31.9%–32.3%), and 31.8% (31.6%–32.0%; $P<0.001$ for Period 2 versus Period 1; $P=0.07$ for Period 3 versus Period 2).

Subgroup analyses are shown in [Tables S8 through S10](#). In the 140–160 mm Hg group (38.2% of the overall population), despite the proportion with SBP <140

mm Hg increasing in most groups between Periods 2 and 3, non-Hispanic Black patients observed a decrease from 69.2% (68.1%–70.2%) to 66.8% (65.8%–67.9%) ($P=0.008$; [Table S9](#)). In non-Hispanic Other patients (ie, multiracial, Asian American, American Indian or Alaska Native, and Native Hawaiian or Other Pacific Islander), SBP control to <140 mm Hg increased from 69.5% (66.7%–72.4%) in Period 2 to 75.0% (72.2%–78.0%) in Period 3 ($P=0.02$), and <130 mm Hg increased from 28.9% (27.5%–30.5%) in Period 1 to 33.7% (31.7%–35.7%) in Period 2 ($P<0.001$).

In Veterans with pretreatment SBP ≥ 160 mm Hg (5.2% of the overall population), in female Veterans, SBP control to <140 mm Hg increased from 56.6%

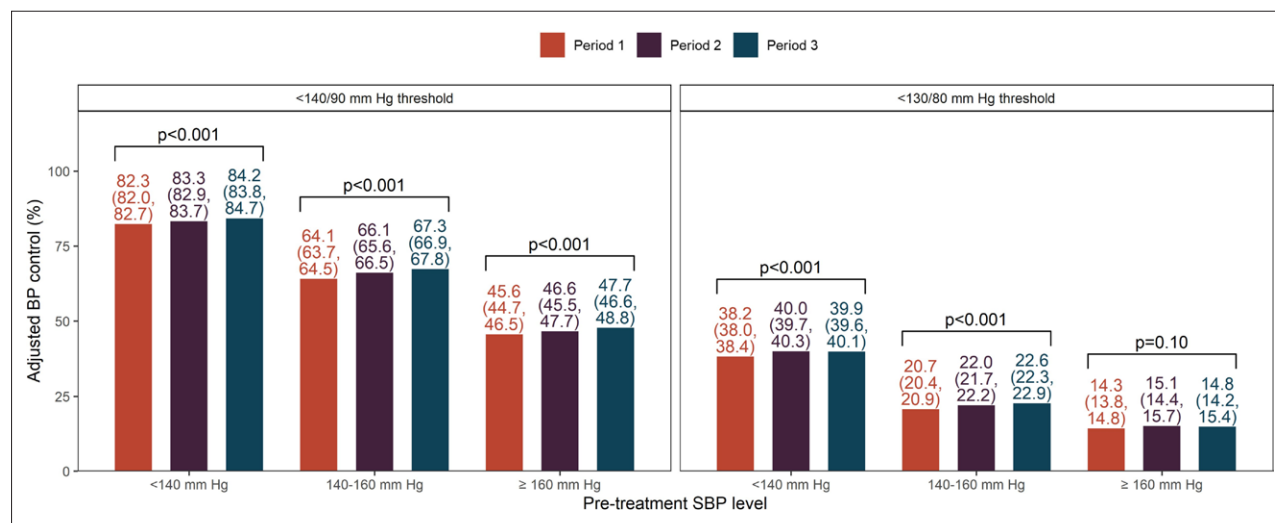


Figure 3. One-year blood pressure (BP) control among Veterans initiating antihypertensive medication.

The figure shows adjusted 1-year posttreatment systolic BP (SBP) control in each period, with 95% CIs contained in parentheses. Values are derived from models adjusting for calendar month, age, sex, race, ethnicity, diabetes, chronic kidney disease, and history of cardiovascular disease.

(52.7%–60.7%) in Period 1 to 64.2% (59.7%–69.0%) in Period 2 ($P=0.04$), before decreasing to 52.1% (48.4%–56.1%) in Period 3 ($P<0.001$; Table S10). A decreasing proportion of female Veterans with posttreatment SBP <130 mmHg occurred from 23.4% (20.8%–26.4%) in Period 2 to 18.6% (16.4%–21.0%) in Period 3 ($P=0.02$). Veterans of Hispanic ethnicity observed a decrease in SBP control to <140 mmHg from 54.7% (49.2%–60.9%) in Period 2 to 44.2% (39.7%–49.3%) in Period 3 ($P=0.02$). Finally, in the non-Hispanic Other group, the mean SBP increased from 140.6 (137.2–144.0) mmHg in Period 1 to 147.2 (143.5–150.8) mmHg in Period 2 ($P=0.04$); correspondingly, the proportion with SBP control to <130 mmHg decreased from 22.5% (18.9%–26.9%) to 12.0% (9.3%–15.5%; $P<0.001$).

VISN-level BP control rates varied by $<5\%$ to 10%, depending on the control definition and period (Figures S8 through S10), and including VISN as a random effect did not change the estimates (Figures S11 and S12). In sensitivity analyses, removing dropout weights did not alter the primary findings, although the 95% CIs were wider (Table S11). Restricting study entry criteria to only 1 valid BP measurement yielded similar results to the primary analysis, with tighter CIs.

DISCUSSION

The current analysis has several important findings. First, among Veterans initiating antihypertensive medication, the pretreatment BP level did not meaningfully change from before to after publication of the 2017 ACC/AHA BP guideline or the COVID-19 pandemic. For patients with stage 2 hypertension in particular (SBP ≥ 140 mmHg), this stability despite lower recommended treatment thresholds is a classic signal of clinical inertia, the failure to initiate or intensify therapy when indicated. Second, the proportion of patients with BP control 1 year after medication initiation also did not change meaningfully during the same time periods, with nominal increases in medication adherence. Third, only half to two-thirds of Veterans with stage 2 hypertension in the most contemporary period of the study had BP controlled to $<140/90$ mmHg, and fewer than a quarter had controlled BP at the more stringent $<130/80$ mmHg threshold. Finally, we observed notable differences in BP control among Veterans who were female or from racial and ethnic minority backgrounds. As the next iteration of the clinical guidelines will begin to inform care in 2025 and beyond,²⁶ these findings underscore the need to match lower diagnostic thresholds with equity-focused, action-oriented treatment algorithms that emphasize earlier initiation of evidence-based treatments, rapid titration, and adherence support to accelerate BP control and close persistent sex- and race/ethnicity-based gaps in cardiovascular outcomes among Veterans. Nonetheless, the 12–25 mmHg mean SBP reductions observed

across pretreatment strata represent clinically important improvements, as prior meta-analyses show that every 10 mmHg reduction in SBP lowers cardiovascular event risk by 20%.⁴² Even when guideline targets are not achieved, the magnitude of these SBP reductions meaningfully reduce future cardiovascular risk and highlights the importance of sustaining and intensifying therapy to maintain long-term control.

The current findings add to a body of literature evaluating the impact of guidelines on BP control. Our findings align with previous work concluding that BP control rates did not largely improve after the guidelines were published in 2017.² Among US adults with prevalent hypertension taking antihypertensive medication in the National Health and Nutrition Examination Surveys 2015 to 2016, 66.7% had controlled BP $<140/90$ mmHg,² with little change in 2017 to 2020 (67.8%). These findings are similar to the current analysis, where BP control to $<140/90$ mmHg among Veterans with pretreatment SBP 140–160 mmHg was 64.1% in Period 1 and 66.1% in Period 2 ($P<0.001$). National Health and Nutrition Examination Surveys data also noted a decreasing trend in BP control ($<140/90$ mmHg) in non-Hispanic Black individuals (58.2% in 2015–2016 to 52.6% in 2017–2020; $P=0.077$), similar to the small decrease noted in the current study during the same period (69.2%–66.8%; $P=0.008$). In the highest-risk SBP stratum (≥ 160 mmHg), the current analysis observed marked increases in SBP control to <140 mmHg after the BP guidelines were published in patients who identify as female (56.6%–64.2%; $P=0.04$) and non-Hispanic Black (46.3%–51.0%; $P=0.03$). In contrast, male Veterans and Veterans of non-Hispanic White race and ethnicity had relatively stable BP control rates between Periods 1 and 2 in all SBP strata. These findings collectively underscore that disparities in BP control by sex, race, and ethnicity persist after publication of the clinical guidelines.⁴³

Although racial disparities in BP control in the general population are attributed to differences in health-care access, the integrated structure of the VHA may minimize the impact of many traditional access barriers (eg, cost of care). Other mechanisms may therefore contribute to the observed race and ethnicity differences in BP control. One potential consideration is that Black adults more commonly exhibit low-renin, salt-sensitive hypertension and may require more intensive multi-drug therapy to achieve BP control.⁴⁴ In the presence of therapeutic inertia, these patients may be less likely to achieve target BP. In addition, factors such as medication acceptance, patient mistrust, perceived discrimination,^{45–47} and broader social and structural influences may affect hypertension management even within integrated healthcare systems such as the VHA.^{48–50}

Extending into the pandemic period, National Health and Nutrition Examination Surveys data noted an overall

BP control rate of 68.3% among those taking antihypertensives in 2021 to 2023 ($P=0.65$ versus 2017–2020), similar to the current analysis (70.1% of the middle SBP stratum with SBP <140 mmHg in Period 3). Among 1.7 million US hypertensive patients from 24 health systems included in the PCORNet BP Control Lab, BP control to <140/90 mmHg in the pre- and postpandemic eras was highly variable among health systems (range, 46%–74%), with an average decrease in BP control of 7.2% (95% CI, 5.4%–9.0%) with the onset of the pandemic ($P<0.001$).⁶ One analysis of 464 585 US adults enrolled in an employer-sponsored wellness program reported a mean increase of 1.1 to 2.5 mmHg during each month of the pandemic period despite stable month-to-month SBP during the prepandemic period.⁷ Another multisystem analysis of 137 593 US adults, BP control decreased by 3.4% during the pandemic.⁸ Unlike these analyses, the current analysis indicated that BP control to <140/90 mmHg stagnated or only slightly increased (max absolute change 2.3%) during COVID-19 in most Veteran groups. The differences between our findings and these prior studies could be due to inherent differences in study populations (ie, treated versus untreated patients) or the current analysis including more postpandemic years (>3) compared with the other studies (max 1), which likely represent the acute peripandemic period. Finally, the current analysis supports and adds to findings from a separate VHA analysis of 618 497 treated hypertensive Veterans with uncontrolled BP in the prepandemic period demonstrating that the pandemic had minimal impact on BP control (risk ratio, 0.98 [95% CI, 0.98–0.99]).¹⁰ Taken together, these findings highlight the importance of ongoing longitudinal surveillance to understand evolving patterns of BP control and to prevent widening disparities as the pandemic's longer-term effects continue to unfold.

In 2012, Fletcher et al¹ reported improvement in BP control rates (<140/90 mmHg) from 45.7% in 2000 to 75.3% in 2010 at 15 VHA facilities. In the sensitivity analyses collapsing pretreatment SBP strata of the current analysis, BP control rates were similar at this threshold (73.7%–75.7%). However, evaluating only those with stage 2 hypertension (ie, pretreatment BP $\geq$ 140 mmHg), the current analysis demonstrated much lower BP control rates at the <140/90 mmHg threshold in the most recent Period (67.3% for the middle SBP stratum and 47.7% for the highest SBP stratum) and even lower rates at the stricter <130/80 mmHg threshold (39.9% for the middle SBP stratum and 14.8% for the highest SBP stratum). These findings serve to provide more contemporary data of new-users of antihypertensive medications and call attention to the low 1-year BP control rates at the stricter <130/80 mmHg threshold, which is the recommended treatment target in the 2025 AHA/ACC guidelines.²⁶ Although the current analysis focused on new-users, the findings are similar to prior

estimates of prevalent users (28% of over 2.7 million Veterans with prevalent hypertension between 2016 and 2017 had BP <130/80 mmHg),⁹ lending some validity to our results. The similar rates between the prevalent user and the current new-user study also indicate an opportunity to optimize treatment after the initial treatment period. Nonetheless, the BP control rates in the current study are close to the Healthy People 2030 goal to have 18.9% of hypertensive adults with controlled BP to <130/80 mmHg,⁵¹ with notable opportunity to improve control rates for those with pretreatment SBP $\geq$ 160 mmHg. Although some Veterans may achieve control after 1 year, delayed titration is clinically important: earlier control and greater time in BP target range are associated with fewer cardiovascular events, independent of baseline BP.^{52–55}

Although the current analysis focuses on patient-level, early BP management, center and provider context matters. Prior work shows measurable between-site variability in VA BP control and prescribing,^{56,57} and that team-based/pharmacist-involved models can improve control,⁵⁸ while provider-level factors contribute to therapeutic inertia.⁵⁹ In the VA, initial treatment decisions vary by provider type,⁶⁰ and resident physicians are 22% to 54% more likely to intensify therapy more often than attending or mid-career providers.⁶¹ Provider-level factors associated with inertia also include clinical workload, competing demands, and uncertainty about when to intensify therapy.^{59,61} The current analysis demonstrates that a majority (75%–80%) of Veterans are initiated on antihypertensive monotherapy, similar to prior VA analyses.⁶⁰ Veterans initiating monotherapy experience a median 2-month interval from medication initiation to first outpatient visit,⁶² double the 1-month follow-up period recommended by the 2017 ACC/AHA guidelines⁴ and 2020 VA/DoD BP guidelines.⁶³ Together, these findings highlight a critical and clinically actionable window of postinitiation therapeutic inertia. In this context, the suboptimal BP control observed at 1 year likely reflects delays in repeat BP evaluation and intensification rather than inadequate initial treatment response alone, highlighting early postinitiation care as a critical window for improving hypertension control.

The VHA became a national leader in hypertension care by coupling evidence-based treatments with system-wide programs to drive their implementation. These programs include VHA-specific clinical guidelines, treatment algorithms, remote patient monitoring, clinical decision support, quality monitoring dashboards, performance measures, audit-and-feedback, and patient-centered medical homes, among others.^{1,64} Although promotion of multidrug regimens helped contribute to improved BP control rates at the VHA in the early 2000s,³⁹ recent data indicate that the reliance on initial monotherapy in the VHA is high (66%) and rising.⁶⁰ In the current study, the initiation of combination therapy

declined from 19.0% in Period 1 to 15.2% in Period 3, with a commensurate decline in use of single-pill combination products from 40.0% to 36.1%. Persisting reliance on monotherapy may both reflect and reinforce inertia: clinicians must first initiate, or switch to, combination therapy to meet <130/80 mmHg targets, yet such intensifications remain uncommon in clinical practice.⁶⁵ To support robust follow-up care, since 2003, the VHA's remote telemonitoring program has enabled Veterans to access a free home BP monitor and frequent follow-up with a care coordinator to discuss readings.⁶⁶ The VHA's rapid telehealth expansion helped to maintain primary care engagement among >1.3 million high-risk Veterans between 2020 and 2022,⁶⁷ which could be one of many potential reasons why we did not observe adverse changes in BP levels during the pandemic period. The VHA also has integrated team-based care into normal workflows via Patient Aligned Care Teams, one component of which includes elevating scopes of practice for pharmacists and nurses to independently manage hypertension.⁶⁸ As hypertension guidelines continue to evolve away from monotherapy and toward intensive targets, as is recommended in the most recent 2025 guideline,²⁶ the VHA must evolve its existing infrastructure to include cross-cutting strategies to accelerate improvement in BP control,⁶⁹ such as efforts to address disparities across subpopulations,⁷⁰ increasing use of single-pill combination products, and implementing clinician-facing interventions to combat inertia.

A major strength of the current study lies in the robust clinical, administrative, and pharmaceutical information available for VHA patients, allowing us to accurately assess demographics, clinical conditions, laboratory measurements, vital signs, and medication use. The VHA provides near equal access to antihypertensive medications for all Veterans via a mail-order pharmacy and preferred 90-day refill system, bolstering sustainable medication use for all Veterans. Because VA clinicians may individualize targets per different guidelines (eg, VA/DoD, ACC/AHA 2017, or ACP/AAFP 2017), we reported multiple thresholds (combined and component SBP/DBP) and added SBP <150 mmHg for ≥60 years to enhance generalizability across practice standards. Nonetheless, this study should be interpreted in the context of several limitations. BP was measured during clinical care and may not have been measured in a standardized manner for Veterans included in the current study. We also did not include home BP measurements, as these data are not stored in the database and require natural language processing to query from free-text clinical notes. However, clinical decisions may often have been based on patient-reported home BP measurements. Given that we sought to isolate impacts of early treatment (a time period that exerts substantial influence on BP control) through a new-user study design, we could not estimate awareness, diagnosis,

and treatment rates. However, our inclusion of a variable indicating the interval between diagnosis and treatment revealed a relatively short period between initial diagnosis and treatment (1 year on the median). As with other VHA studies, the population was mostly male and non-Hispanic White, and results may differ in periods with more women or racial and ethnic minority groups. We lacked direct measures of some social determinants of health (eg, income, education, transportation), so residual confounding is possible, and we could not evaluate whether socioeconomic factors modify observed differences in BP control across SDoH subgroups. However, population-wide analyses indicate that pandemic-era declines in BP control were largely attributable to longer intervals between follow-up visits and fewer BP measurements,^{6,10} supporting our emphasis on observed care-continuity pathways while recognizing unmeasured social determinants of health may contribute to these results. Because of the large sample size, some statistically significant differences between periods were very small in magnitude and did not represent clinically meaningful differences. Further, we did not model multilevel (facility/provider) effects in this analysis to focus on patient-level data; as a result, we did not formally quantify regional or local variation in care delivery during the pandemic, although exploratory analyses indicated modest heterogeneity in control rates across VISNs. Finally, we did not know what BP target treating clinicians were trying to achieve and therefore could not directly measure clinical inertia in this large, secondary data set, which is an area of ongoing research.

CONCLUSIONS

In the current study of US Veterans newly diagnosed and treated for hypertension, neither the release of the 2017 ACC/AHA BP guidelines nor the COVID-19 pandemic meaningfully changed pre- or posttreatment BP levels or BP control rates, and variation by sex, race, and ethnicity remains. Only a minority of Veterans reach a stringent guideline-recommended BP goal of <130/80 mmHg within a year of initiating treatment. Despite the VHA's comprehensive system for hypertension care, these findings signal stalled progress in light of the VHA's earlier gains as highlighted by prior studies, particularly for those with stage 2 hypertension, and reveal persistent equity gaps. Reversing this stasis will require re-calibration toward evidence-based interventions explicitly designed to combat clinical inertia and support adherence, such as algorithmic titration, pharmacist-led BP protocols, single-pill combinations, and performance-feedback dashboards.

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